Answers learning task 6

2026 – 2027

Exercise 4.51: Assume that Z has a standard normal distribution. Determine the value for z that solves each of the following:

(a) $P(Z < z) = 0.9$
(b) $P(Z < z) = 0.5$
(c) $P(Z > z) = 0.1$
(d) $P(Z > z) = 0.9$
(e) $P(-1.28 < Z < z) = 0.8$

Answer

We can determine these boundary values z using the graphical calculator using either the function "invnorm" (using Tail: Left if not specified) or the "numeric solver":

(a) $P(Z < z) = 0.9$ holds for $z = \text{InvNorm}(\text{area} = 0.9, \mu = 0, \sigma = 1) \approx 1.2816$.

(b) $P(Z < z) = 0.5$ holds for $z = \text{InvNorm}(\text{area} = 0.5, \mu = 0, \sigma = 1) = 0$ (notice that the standard normal distribution is symmetric in $x = 0$).

(c) $P(Z > z) = 0.1$ holds for $z = \text{InvNorm}(\text{area} = 0.1, \mu = 0, \sigma = 1, \text{Tail} = \text{Right}) \approx 1.2816$.

(d) $P(Z > z) = 0.9$ holds for $z = \text{InvNorm}(\text{area} = 0.9, \mu = 0, \sigma = 1, \text{Tail} = \text{Right}) \approx -1.2816$.

(e) We can rewrite this equation as follows:

$$P(Z < z) - P(Z < -1.28) = 0.8 \Rightarrow P(Z < z) = 0.8 + P(Z < -1.28)$$

This gives us

$$P(Z < z) = 0.8 + \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = -1.28, \mu = 0, \sigma = 1) \approx 0.9003$$
$$\Rightarrow z = \text{InvNorm}(\text{area} = 0.9003, \mu = 0, \sigma = 1) \approx 1.2831.$$

Exercise 4.55: Assume that X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the following:

(a) $P(X < 11)$
(b) $P(X > 2)$
(c) $P(3 < X < 7)$
(d) $P(-2 < X < 9)$
(e) $P(2 < X < 8)$

Answer

The following probabilities can be calculated using the function "normalcdf" on the graphical calculator:

(a) $P(X < 11) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 11, \mu = 5, \sigma = 4) \approx 0.9332$.

(b) $P(X > 2) = \text{normalcdf}(\text{lower} = 2, \text{upper} = 10^{99}, \mu = 5, \sigma = 4) \approx 0.7734$.

(c) $P(3 < X < 7) = \text{normalcdf}(\text{lower} = 3, \text{upper} = 7, \mu = 5, \sigma = 4) \approx 0.3829$.

(d) $P(-2 < X < 9) = \text{normalcdf}(\text{lower} = -2, \text{upper} = 9, \mu = 5, \sigma = 4) \approx 0.8013$.

(e) $P(2 < X < 8) = \text{normalcdf}(\text{lower} = 2, \text{upper} = 8, \mu = 5, \sigma = 4) \approx 0.5467$.

Exercise 4.56: Assume that X is normally distributed with a mean of 5 and a standard deviation of 4. Determine the value for x that solves each of the following:

(a) $P(X > x) = 0.5$
(b) $P(X > x) = 0.95$
(c) $P(x < X < 7) = 0.2$
(d) $P(3 < X < x) = 0.95$
(e) $P(-x < X - 5 < x) = 0.99$

Answer

We can determine these boundary values x using the graphical calculator using either the function "invnorm" (using Tail: Left if not specified) or the "numeric solver":

(a) $P(X > x) = 0.5$ holds for x equal to the mean, so $x = 5$.

(b) $P(X > x) = 0.95$ holds for $x = \text{InvNorm}(\text{area} = 0.95, \mu = 5, \sigma = 4, \text{Tail} = \text{Right}) \approx -1.5794$.

(c) We can rewrite this equation as follows:

$$P(X < 7) - P(X < x) = 0.2 \Rightarrow P(X < x) = P(X < 7) - 0.2.$$

This gives us

$$P(X < x) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 7, \mu = 5, \sigma = 4) - 0.2 \approx 0.4915.$$
$$\Rightarrow z = \text{InvNorm}(\text{area} = 0.4915, \mu = 5, \sigma = 4) \approx 4.9144.$$

(d) We can rewrite this equation as follows:

$$P(X < x) - P(X < 3) = 0.95 \Rightarrow P(X < x) = 0.95 + P(X < 3).$$

This gives us

$$P(X < x) = 0.95 + \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 3, \mu = 5, \sigma = 4) \approx 1.2585.$$

Notice that this equation is ill-defined, as a probability cannot be larger than one.

(e) $P(Z > z) = 0.9$ holds for $z = \text{InvNorm}(\text{area} = 0.9, \mu = 0, \sigma = 1, \text{Tail} = \text{Right}) \approx -1.2816$.

(f) Notice that $X - 5$ is normally distributed with mean 0 and variance 4. In that case, $-x$ should be at the 0.5-th percentile of the distribution and x at the 99.5-th percentile:

$$x = \text{InvNorm}(\text{area} = 0.995, \mu = 0, \sigma = 4) \approx 10.3033.$$

Exercise 4.63: The fill volume of an automated filling machine used for filling cans of carbonated beverage is normally distributed with a mean of x cc and a standard deviation of 5 cc.

(a) At what value should the mean be set so that $99.9\,\%$ of all cans exceed 365 cc?

(b) At what value should the mean be set so that $99.9\,\%$ of all cans exceed 365 cc if the standard deviation can be reduced to 2 cc?

Answer

The fill volume X is normally distributed with mean x and standard deviation 5.

(a) We want to solve for x the equation $P(X > 365) = 0.999$. As the mean is unknown, we need to use the numeric solver for this:

$$E_1 = \text{normalcdf}(\text{lower} = 365, \text{upper} = 10^{99}, \mu = X, \sigma = 5)$$
$$E_2 = 0.995$$

The numeric solver gives us a mean $x \approx 380.4513$.

(b) We want to solve for x the equation $P(X > 365) = 0.999$. As the mean is unknown, we need to use the numeric solver for this:

$$E_1 = \text{normalcdf}(\text{lower} = 365, \text{upper} = 10^{99}, \mu = X, \sigma = 2)$$
$$E_2 = 0.995$$

The numeric solver gives us a mean $x \approx 371.1805$.

Exercise 4.69: The life of a semiconductor laser at a constant power is normally distributed with a mean of 7000 hours and a standard deviation of 600 hours.

(a) What is the probability that a laser fails before 6000 hours?

(b) What is the life in hours that $95\,\%$ of the lasers exceed?

(c) If three lasers are used in a product and they are assumed to fail independently, what is the probability that all three are still operating after 7000 hours?

Answer

(a) The probability that a laser fails before 6000 hours is equal to

$$P(X < 6000) = \text{normalcdf}(\text{lower} = -10^{99}, \text{upper} = 6000, \mu = 7000, \sigma = 600)$$
$$\approx 0.0478.$$

(b) The life in hours that $95\,\%$ of the lasers exceed is the value x for which $P(X > x) = 0.95$. We can find x by using the "invnorm" function:

$$x = \text{InvNorm}(\text{area} = 0.95, \mu = 7000, \sigma = 600, \text{Tail} = \text{Right}) \approx 6013$$

(c) The probability that a single laser still operates after 7000 hours is equal to

$$P(X > 7000) = \text{normalcdf}(\text{lower} = 7000, \text{upper} = 10^{99}, \mu = 7000, \sigma = 600) = 0.5.$$

Then, if we assume all three lasers fail independently, the probability that all three still operates after 7000 hours is $0.5 \cdot 0.5 \cdot 0.5 = 0.125$.

Exercise 4.91: Suppose X has an exponential distribution with mean equal to 10. Determine the following:

(a) $P(X > 10)$ (b) $P(X > 20)$ (c) $P(X < 30)$
(d) Find the value of x such that $P(X < x) = 0.90$.

Answer

For a random variable X that is exponentially distributed with mean 10 (i.e., parameter $\lambda = \frac{1}{10}$), we have that

$$P(X > x) = e^{-\lambda x} = e^{-\frac{1}{10}x}$$

We can calculate the desired probabilities given this formula:

(a) $P(X > 10) = e^{-\frac{1}{10}\cdot 10} = e^{-1} \approx 0.3679$.

(b) $P(X > 20) = e^{-\frac{1}{10}\cdot 20} = e^{-2} \approx 0.1353$.

(c) $P(X < 30) = 1 - P(X > 30) = 1 - e^{-\frac{1}{10}\cdot 30} = 1 - e^{-3} \approx 0.9502$.

(d) $P(X < x) = 0.90$ means $P(X > x) = e^{-\frac{1}{10}\cdot x} = 0.10$. Therefore $-\frac{1}{10} \cdot x = \ln 0.10 = -\ln 10 \Rightarrow x = 10 \ln 10 \approx 23.0259$.

Exercise 4.103: The lifetime of a mechanical assembly in a vibration test is exponentially distributed with a mean of 400 hours.

(a) What is the probability that an assembly on test fails in less than 200 hours?

(b) What is the probability that an assembly operates for more than 500 hours before failure?

(c) If an assembly has been on test for 400 hours without a failure, what is the probability of a failure in the next 100 hours?

(d) If 10 assemblies are tested, what is the probability that at least one fails in less than 100 hours? Assume that the assemblies fail independently.

(e) If 10 assemblies are tested, what is the probability that all have failed by 800 hours? Assume that the assemblies fail independently.

Answer

For a random variable X that is exponentially distributed with mean 400 (i.e., parameter $\lambda = \frac{1}{400}$), we have that

$$P(X > x) = e^{-\lambda x} = e^{-\frac{1}{400}x}$$

We can calculate the desired probabilities given this formula:

(a) $P(X < 200) = 1 - P(X > 200) = 1 - e^{-\frac{1}{400}\cdot 200} = 1 - e^{-1/2} \approx 0.3935$.

(b) $P(X > 500) = e^{-\frac{1}{400}\cdot 500} = e^{-5/4} \approx 0.2865$.

(c) As the exponential distribution has the memoryless property, this probability is just

$$P(X < 100) = 1 - P(X > 100) = 1 - e^{-\frac{1}{400}\cdot 100} = 1 - e^{-1/4} \approx 0.2212.$$

(d) Introduce a new variable counting the number of failing assemblies out of 10 within 100 hours. Then Y is binomially distributed with $n = 10$ and $p = P(X < 100) = 0.2212$. The probability that at least one fails is Then

$$P(Y \geq 1) = 1 - P(Y = 0) = 1 - \text{binompdf}(n = 10, p = 0.2212, k = 0) \approx 0.9179.$$

(e) Introduce a new variable counting the number of failing assemblies out of 10 within 100 hours. Then Y is binomially distributed with $n = 10$ and $p = P(X < 800) = 1 - e^{-\frac{1}{400} \cdot 800} = 1 - e^{-2} \approx 0.8647$. The probability that all have failed by 800 hours:

$$P(Y = 10) = \text{binompdf}(n = 10, p = 0.8647, k = 10) \approx 0.2336.$$

Exercise 4.165: Suppose that $f(x) = 0.5x - 1$ for $2 < x < 4$. Determine the following:

(a) $P(X < 2.5)$ (b) $P(X > 3)$ (c) $P(2.5 < X < 3.5)$
(d) Determine the cumulative distribution function of the random variable.
(e) Determine the mean and variance of the random variable.

Answer

We start with taking the antiderivative of $f(x)$ to obtain the cumulative distribution function (CDF):

$$\begin{aligned} P(X \leq x) = \int_{y=2}^{x} 0.5y - 1 \, \mathrm{d}y &= \left[0.25y^2 - y\right|_{y=2}^{x} \\ &= \left(0.25x^2 - x\right) - (1 - 2) \\ &= 0.25x^2 - x + 1 \end{aligned}$$

The cumulative distribution function (question (d)) $F(x)$ can then be given as

$$F(x) = \begin{cases} 0, & \text{for } x \leq 2 \\ 0.25x^2 - x + 1, & \text{for } 2 < x < 4 \\ 1, & \text{for } x \geq 4 \end{cases}$$

We can use this CDF to compute the desired probabilities:

(a) $P(X < 2.5) = P(X \leq 2.5) = F(2.5) = 0.25 \cdot (2.5)^2 - 2.5 + 1 = 0.0625$

(b) $P(X > 3) = 1 - P(X \leq 3) = 1 - F(3) = 1 - 0.25 \cdot 3^2 - 3 + 1 = 0.75$

(c)

$$\begin{aligned} P(2.5 < X < 3.5) &= P(X \leq 3.5) - P(X \leq 2.5) \\ &= F(3.5) - F(2.5) \\ &= \left(0.25 \cdot (3.5)^2 - 3.5 + 1\right) - \left(0.25 \cdot (2.5)^2 - 2.5 + 1\right) \\ &= 0.5 \end{aligned}$$

(e) We can compute the mean of this random variable as follows:

$$\begin{aligned} E(X) = \int_{x=-\infty}^{\infty} x \cdot f(x)\,\mathrm{d}x &= \int_{x=2}^{4} x \cdot (0.5x - 1)\,\mathrm{d}x \\ &= \int_{x=2}^{4} \cdot 0.5x^2 - x\,\mathrm{d}x \\ &= \left[\frac{1}{6}x^3 - \frac{1}{2}x^2\right|_{x=2}^{4} \\ &= \left(\frac{32}{3} - 8\right) - \left(\frac{4}{3} - 2\right) \\ &= \frac{10}{3}. \end{aligned}$$

For the variance, we first compute $E(X^2)$:

$$\begin{aligned} E(X^2) = \int_{x=-\infty}^{\infty} x^2 \cdot f(x)\,\mathrm{d}x &= \int_{x=2}^{4} x^2 \cdot (0.5x - 1)\,\mathrm{d}x \\ &= \int_{x=2}^{4} 0.5x^3 - x^2\,\mathrm{d}x \\ &= \left[\frac{1}{8}x^4 - \frac{1}{3}x^3\right|_{x=2}^{4} \\ &= \left(32 - \frac{64}{3}\right) - \left(2 - \frac{8}{3}\right) \\ &= \frac{34}{3}. \end{aligned}$$

We now use the identity $V(X) = E(X^2) - (E(X))^2$:

$$V(X) = E(X^2) - (E(X))^2 = \frac{34}{3} - \left(\frac{10}{3}\right)^2 = \frac{2}{9}.$$